

Narrative Transportation in Gamified Information Systems: The Role of Narrative-Task Congruence

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ABSTRACT

Gamification refers to the use of game design elements in non-game contexts to evoke both instrumental and experiential outcomes within users. Although narratives are an integral part of most video games, little academic attention has been paid toward narratives as part of gamification design. In this research, we investigate two distinct aspects of narratives in gamified information systems (IS): (1) the mental state of narrative transportation, and (2) the congruence between a narrative and an instrumental task. In an online experiment in the context of physical activity among 325 individuals, we find that high levels of narrative-task congruence make it easier for users to experience narrative transportation, which in turn positively influences the dual outcomes of gamified IS. We contribute to research by contextualizing narrative transportation theory and empirically investigating the affective and behavioral effects of congruence between instrumental task and game design elements.

CCS CONCEPTS

• **Information systems** → Information systems applications.

KEYWORDS

Gamification, Narrative, Online experiment, Gamified information systems

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1 INTRODUCTION

Designers of gamified information systems (IS) use game design elements to engage users and provide them with enjoyable experiences while simultaneously fostering instrumental task outcomes.

This duality of desired outcomes, called meaningful engagement, distinguishes gamified IS from traditional IS that are designed in a way which mostly ignores experiential outcomes [1]. Ever since gamification emerged in the early 2010s, scholars have focused their attention on a specific set of elements including points, goals, badges, or leaderboards [2]. Little scholarly attention has been paid toward other game design elements, like narratives, as part of gamification design [3, 4]. Yet, extant research in disciplines such as marketing [5], politics [6], education [7], and healthcare [8] has demonstrated that narratives host immense persuasive potential. Numerous studies show that when individuals recognize a coherent narrative within received information, this may lead to changes in affect, cognition, attitudes, intentions, and even actual behavior [9]. Narratives can also be highly entertaining and engaging [10]. Why else can we sometimes not stop watching our favorite Netflix series or are unable to put down a compelling novel?

In practice, various examples demonstrate how well-crafted narratives in gamified IS can contribute to their success. For instance, *Zombies, Run!*, an app regularly hailed as a prime example for good gamification design, uses audio plays to immerse users in a zombie apocalypse narrative to motivate users to engage in physical exercise [11]. Other examples of narratives in gamified IS include the language education app Duolingo, which uses textual and audio-based mini-stories to challenge users' reading and listening comprehension, or the Microsoft Office add-on Ribbon Hero, which introduces users to new features based on the story of Clippy the paperclip [1].

The potential to combine persuasion with entertainment makes narratives, as a game design element, a prime candidate to support the dual outcomes of gamification. However, many fundamental questions regarding the effects of narratives in gamified IS remain unanswered. For example, it is not yet clear under which circumstances narratives engender the same level of engagement in non-game contexts as they do in video games [1]. Accordingly, it remains largely unclear how to leverage narratives in gamified IS to evoke meaningful engagement of users.

In this work, we take a step toward overcoming this lack of knowledge and shine light onto the effects of narratives in gamified IS. We draw on narrative transportation theory [12, 13], which is a prevalent theoretical lens for studying the affective, cognitive, and behavioral effects of narratives in many disciplines. The theory postulates that when individuals lose themselves in a narrative world, they enter a mental state of narrative transportation, which

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explains the persuasive effects of narratives [9, 13]. Besides investigating the general effects of narrative transportation on meaningful engagement, we also focus on the interactions between the characteristics of instrumental task and narrative. Extant gamification research has started to discuss that game design elements must be congruent with the targeted instrumental task to be more effective (i.e., tasks and game design elements must be compatible with each other) [1]. However, it remains unclear how different levels of congruence between instrumental task and game design element influence individuals' perceptions of game design elements and how they manifest in experiential or instrumental outcomes. Therefore, we ask the following research questions (RQs) in our study:

RQ1: How does narrative transportation influence the dual outcomes desired in gamified IS?

RQ2: What is the role of narrative-task congruence in evoking narrative transportation within users of gamified IS?

To address our RQs, we conducted an online experiment among 325 individuals. As a research context, we chose a gamified mobile application for physical activity (e.g., *Zombies, Run!*). In practice, narratives in such applications have shown to be particularly suitable for promoting instrumental task outcomes. Furthermore, physical activity is one of the most prevalent application areas of gamification [14, 15]. Our results indicate that congruence between instrumental task and characteristics of the narrative can play an important role in determining whether users of gamified IS experience narrative transportation and that users of gamified IS enjoy using the system more when they experience narrative transportation. We contribute to gamification research by opening the academic discussion on the role of narrative transportation in evoking meaningful engagement within users of gamified IS and answer the call for more theory-driven research explaining the behavioral effects of single game design elements thereby moving beyond classical elements of gamification research such as points, badges, and leaderboards [2].

The remainder of this paper is structured as follows. In the second chapter, we provide the theoretical background on narratives in gamified IS and narrative transportation. In chapter three, we derive our hypotheses, before we describe the research context and data collection methods of our study in chapter four. In chapter five we outline the results of our study. In chapter six we discuss the main findings and implications of our research. Finally, we briefly conclude our late breaking work in chapter seven.

2 THEORETICAL BACKGROUND

2.1 Narratives in Gamified Information Systems

Gamification is “the intentional use of game [design] elements for a gameful experience of non-game tasks and contexts” [16]. Gamified IS have a dual purpose of evoking both instrumental and experiential outcomes, the achievement of which is called meaningful engagement [1]. Fitting experiential outcomes in well-designed gamification are influenced by the specific implemented game design elements. While leaderboards may be suitable to foster feelings of social comparison, narratives can be effective in enabling immersion [2]. Thereby, enhanced experiential outcomes must not just be an added benefit for gamified IS [1]. Instead, they should be

associated with higher levels of instrumental outcomes to achieve meaningful engagement [17]. If this is not the case, users may only focus on what affords entertainment and bypass features that are related to instrumental outcomes [18, 19].

A recent review by Koivisto and Hamari [2] concluded that gamification research is dominated by progress and achievement-oriented game design elements (e.g., points, badges, leaderboards), whereas immersion-related elements such as narratives and avatars are much less frequently studied. Some researchers have started investigating outcomes of narratives in gamified IS, which revealed mixed effects. For example, Seo et al. [20] concluded that narratives embedded in gamified IS are effective in positively influencing intrinsic motivation and affective responses in car manufacturing workers, and Brandy [21] shows that the engagement with stories in the mobile language learning application Duolingo results in an increase of implicit and explicit knowledge. Furthermore, Li et al. [22] show that narratives yield positive effects on behavioral intentions of mobile fitness app users. In contrast, Armstrong and Landers [23] found that the use of narratives in gamified IS for information security education increased user satisfaction, but reduced training effectiveness. While these studies provide valuable first insights into the persuasive effects that narratives in gamified IS might hold, more can be investigated regarding the mental state of narrative transportation, its gamification-specific antecedents, and its influence on meaningful engagement. This work aims to overcome this gap by understanding how, and under which circumstances, narratives in gamified IS result in desired behavioral outcomes.

2.2 Narrative Transportation

Narrative transportation refers to a phenomenon in which individuals lose themselves in a story and mentally enter a world that said story evokes [9]. As proposed in the transportation-imagery model [13, 24], certain preconditions must be met for narrative transportation to occur. These preconditions can roughly be classified into characteristics of the narrative and characteristics of the narrative receiver [9]. For characteristics of the narrative, extant research has mainly focused on three factors: (1) identifiable characters, (2) imaginable plot, and (3) verisimilitude [9]. More recently, research has started to reveal additional antecedents, like whether narratives on social media are user-generated [25], the chosen medium through which narratives are brought to the audience [26], or the quality of the narrative on various dimensions, such as realism or narrative environment [27].

In this research, we understand narratives as a game design element that provides information about a story plot including its world, characters, and events via written text or any other audio-visual medium so that it can be interpreted by a receiver [9]. Further, we divide narratives into two main components [28]: (1) narrative content (i.e., what is told) and (2) narrative discourse (i.e., how it is told). Narrative content refers to the totality of characters, actions, and events taken by themselves [29]. It “reflects the linear sequence of events as characters live through them, that is, the backbone and structure describing who did what, where, when, and why” [30]. In contrast, narrative discourse refers to how the story is told as it manifests in a written text, audio-visual medium, or any other type

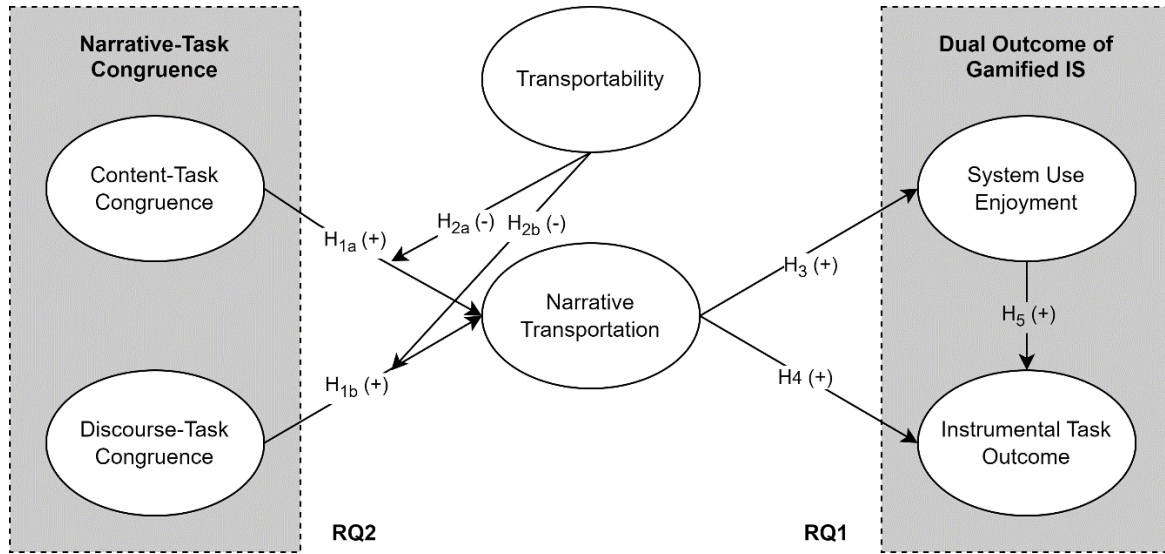


Figure 1: Research Model

of narrative object [9, 31]. It comprises many aspects, including the order of events in which they are told, the voice and perspective of the narrator, or the frequency in which events are told to emphasize their importance [28, 29]. Narrative content and narrative discourse both play an important role in evoking narrative transportation. Recently, Van Laer et al. [30] demonstrated that different dimensions of narrative content and narrative discourse influence the extent to which individuals experience narrative transportation while reading consumer travel reviews on Tripadvisor.

3 RESEARCH MODEL AND HYPOTHESES

In this study we aim to understand narratives in gamified IS and their influence on users' instrumental task outcome through the mental state of narrative transportation. To develop our research hypotheses, we draw on literature from narrative transportation theory [12], the transportation-imagery model [13] and meaningful engagement in gamified IS [1]. Based on the hypotheses, we have conceptualized our research model as depicted in Figure 1.

When individuals aim to extract meaning from a narrative, they construct mental models of the situations and characters described to knit together the causal relations amongst them [32]. If inconsistencies within the narrative become too blatant, individuals have a hard time to comprehend the narrative as it becomes difficult for them to incorporate new information into the existing mental models [33]. In the context of TV advertisements, Glaser and Reisinger [34] demonstrated that ads with strong links between narrative and product lead to higher narrative transportation. We argue that much like a product can become part of a narrative in a TV ad, the instrumental task of a gamified IS can become a part of a narrative in a gamified IS. For example, in *Zombies, Run!*, events (e.g., zombies chasing after the user's character) are used to trigger a specific activity (e.g., sprinting) by the user.

Here, we propose narrative-task congruence as an important characteristic of narratives in gamified IS that influences users'

narrative transportation. We refer to narrative-task congruence as the extent to which a narrative is compatible with the instrumental task of a gamified IS. A low congruence between task and narrative can constitute inconsistencies in the mental model that a user produces of a narrative [32], which leads to a loss of transportation [34]. Narrative-task congruence can occur for both main components of narratives. For example, the narrative content of a zombie apocalypse may be highly congruent with the instrumental task of completing running exercises but be narrated with narrative discourse that does not fit a running exercise (e.g., in a calm and steady voice). Accordingly, we hypothesize:

H1a: *Congruence on the content-task dimension will lead to more narrative transportation within users.*

H1b: *Congruence on the discourse-task dimension will lead to more narrative transportation within users.*

Transportability refers to an individual's chronic propensity to be transported by a narrative, which functions independently of any particular narrative or genre [9, 35]. Users who exhibit high transportability experience fewer barriers toward narrative transportation. They are less predisposed to look for inconsistencies within narratives and it is, thus, easier for them to construct mental models despite incoherent information [33, 36]. As argued before, low levels of congruence between narrative and instrumental task in gamified IS constitute such inconsistencies that prevent individuals from forming mental models and experience narrative transportation. For individuals with high levels of transportability, it is more likely than for others that they do not feel disturbed by these inconsistencies or do not even perceive them as such while receiving the narrative. Thus, they are more likely to enter the mental state of narrative transportation despite low levels of narrative-task congruence. We hypothesize:

H2a: *A user's transportability negatively moderates the relationship between content-task congruence and narrative transportation.*

H2b: *A user's transportability negatively moderates the relationship between discourse-task congruence and narrative transportation.*

Individuals enjoy narratives for a multitude of reasons, including a desire to be entertained, or to experience suspended reality [9, 37]. Many studies have found positive relationships between narrative transportation and positive affective responses [37, 38]. However, the enjoyment of a transportation experience does not necessarily lie in positive emotions evoked by a narrative, but in the process of temporarily leaving one's reality behind [10]. For example, many individuals enjoy being transported into the worlds of horror movies despite them evoking unpleasant emotions, such as fear, or anxiety [10, 39]. We argue that regardless of the narrative content and the narrative discourse itself, users of gamified IS feel more positive about their interactions with the system when they experience high levels of narrative transportation. We hypothesize:

H3: *High levels of narrative transportation will lead to higher levels of system use enjoyment.*

Narrative transportation theory postulates that when individuals experience narrative transportation, their beliefs, attitudes, and intentions tend to change to reflect those embedded in the received narrative [9]. When an individual has sufficient ability to perform a particular behavior, and there are no environmental constraints, these attitudes, beliefs, and intentions may translate to changes toward narrative-consistent behavior [9, 40]. One reason for this is that individuals who experience high levels of narrative transportation tend to have reduced psychological reactance toward promoted behaviors [41]. First studies demonstrate the potential of narrative transportation to support behaviors like smoking cessation [42] or food consumption [43]. In the context of gamified IS we argue that when users experience high levels of narrative transportation into a narrative world that encourages them to perform certain instrumental tasks, their behaviors will tend to align accordingly. Thus, we hypothesize:

H4: *High levels of narrative transportation will lead to higher levels of instrumental task outcome.*

In IS research, a positive relationship between system use enjoyment and instrumental task outcomes is well documented [44, 45]. Research has found that when users enjoy system use, they are more likely to engage with said system and perform its instrumental tasks sufficiently [46, 47]. This is particularly true for IS with hedonic components, where enjoyment is an important antecedent of technology acceptance as users expect the system to be fun and enjoyable to use [48]. Thus, if a user does not enjoy using a gamified IS, they may stop using the system, which would ultimately result in a complete loss of instrumental functionality for this user. In contrast, when individuals perceive higher levels of enjoyment while interacting with the gamified IS they are more likely to stay engaged with the system and perform the instrumental task that the gamified IS targets. We hypothesize:

H5: *High levels of system use enjoyment will lead to higher levels of instrumental task outcome.*

4 STUDY DESIGN

4.1 Research Context

As a research context, we chose a gamified mobile application to support physical activity, since applications like *Zombies*, *Run!* or

Fit for Battle have provided anecdotal evidence on the effectiveness of narratives in fostering physical activity behavior [11]. One reason is that physical activity tasks can easily be implemented into narratives by letting characters within the story perform similar actions to those that are desired to be performed by users [49]. As existing applications do not provide the sufficient variety of content-task congruence and discourse-task congruence that we require for our study, we decided to design mockups and audio narratives for our own application called *Run, Robin!*. In *Run, Robin!* users can listen through a series of audio stories while performing home workouts. While doing so, users accompany Robin Hood on his missions, robbing from the rich and giving to the poor. Audios include interval training and ask users to run in place at their regular pace or to sprint in place. During workout sessions, a voice over provides clear workout instructions (i.e., "start running", "regular pace", "sprint", and "run completed").

4.2 Data Collection

To test our hypotheses, we employed a scenario-based experimental approach. Data was collected with an online survey from a sample of the general US population. A remote online experiment was deemed suitable because it most closely replicated the intended usage of the mobile application (i.e., at home and without observation). Participants were recruited via consumer panels with the help of the market research agency CINT A.B. We applied ex ante recommendations to control for common method bias [50]. Furthermore, we included multiple attention check questions to remove responses of participants who were not reading the questions properly [51]. For both content-task congruence and discourse-task congruence we developed high and low conditions which led to a 2x2 experimental design with a randomized between-subjects arrangement. For the high content-task congruence condition we developed a story in which Robin Hood was running from guards that caught him stealing, while the low content-task congruence told a story of Robin training to shoot with bow and arrow (see A.1 for audio scripts). For the high discourse-task congruence condition we used upbeat, encouraging music and asked the voice actor playing Robin to speak in an exciting and upbeat tone. For the low discourse-task congruence condition, we omitted the music and asked the voice actor to speak calmly and monotonously. Voice acting and music were augmented with fitting sound effects (e.g., footsteps, doors, or forests) to support immersion. Sound effects were held consistent across both discourse-task congruence conditions.

After reading high-level study instructions and answering transportability items, participants were randomized to one of the four experimental groups. We provided participants with a general introduction of *Run, Robin!* including two mockups (see Figure 2). After participants familiarized themselves with the scenario, they were asked to perform a home workout session based on one of the four audio stories that included running and sprinting in place. To ensure that all participants were able to hear the narrative, we conducted a test of their audio devices by asking them to enter a specific code that they received via audio. Furthermore, we asked them to answer open questions to ensure that they were able to comprehend English. All versions of the audio story contained the same workout instructions recorded from a female voice actor in

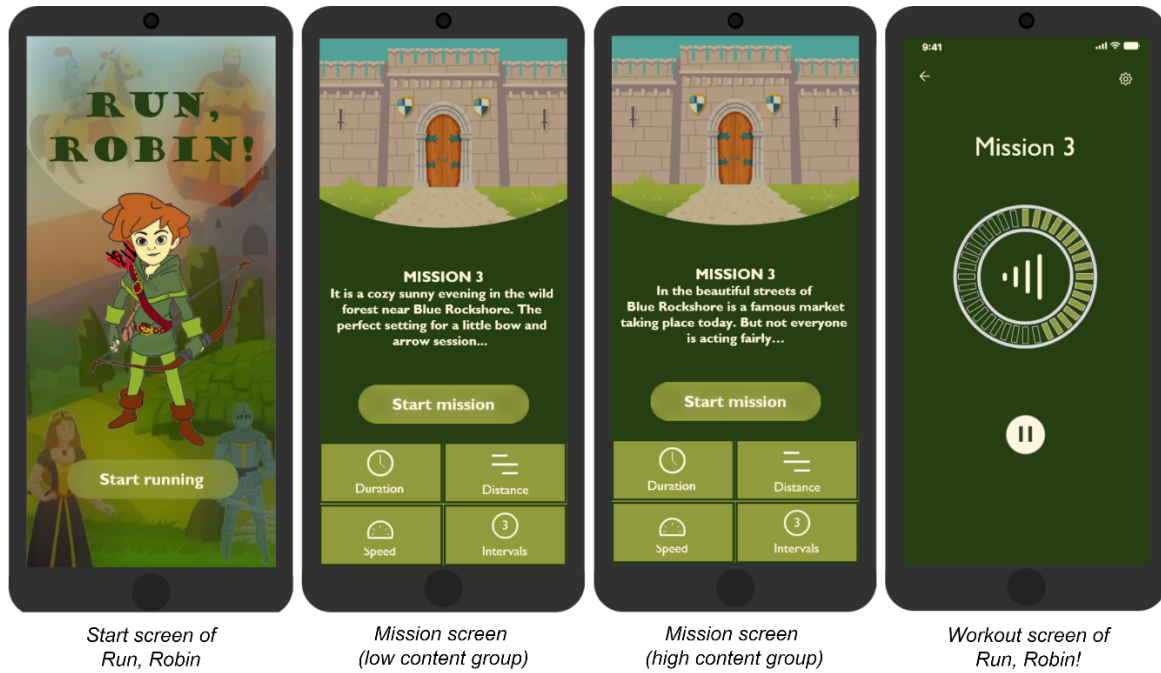


Figure 2: Mockups of Run, Robin!

a neutral tone. Workout sessions were between 4:51 min and 6:39 min long with 3 sprinting intervals each.

After the workout session, we measured manipulation checks, post-experimental values of narrative transportation, system use enjoyment, instrumental task performance as well as additional controls and demographic information. Whenever possible, we used existing scales from literature (see A.2). Due to the novelty of the research topic we had to develop items for content-task congruence, discourse-task congruence, and instrumental task outcome by ourselves. To this end, we developed five initial items for each construct and conducted an item sorting task [52] with 14 researchers that were familiar with similar survey research. During the item sorting task, researchers had to assign each item to one of the constructs and we asked them to provide feedback whether assignments were ambiguous or not. Based on the feedback, the wording of some items was refined and iteratively discussed in the author team. All constructs were measured using a 7-Point Likert scale, from 1 (strongly disagree) to 7 (strongly agree).

5 RESULTS

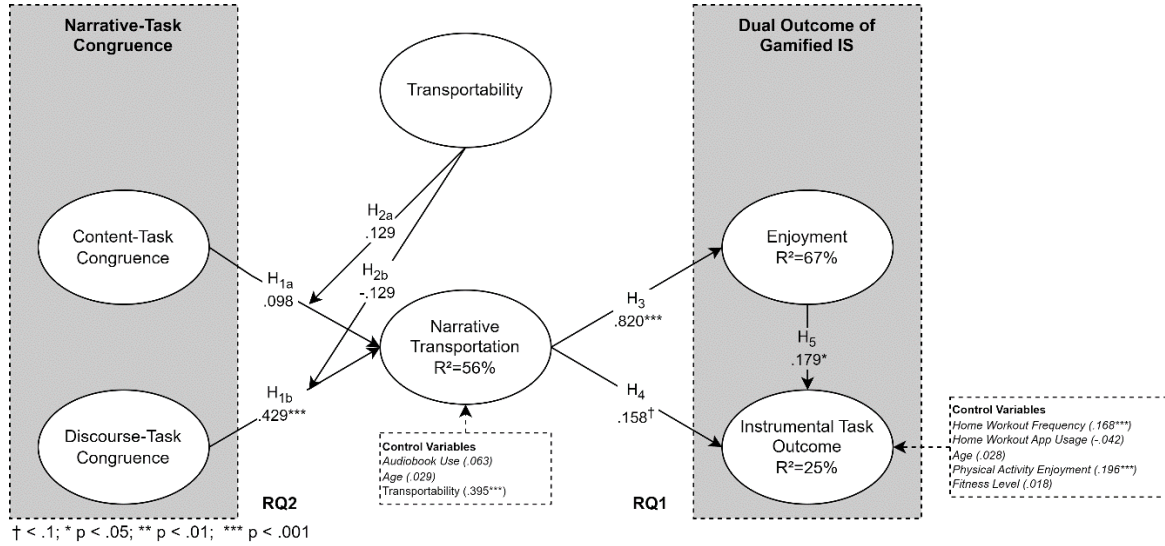
We followed guidelines to maximize data quality for online experiments [51]. Initially, 2,665 individuals clicked on the study link, but most quit after reading the survey instructions. Overall, we obtained 471 full responses. However, we excluded 99 due to duplicate answers, 21 due to rushing the survey, 19 due to straight lining, and 7 due to illogical responses (e.g., participants stating to not have engaged in the activity at all), which left us with 325 valid responses (see A.3). We used the survey results to validate the construct measurement scales and the proposed theoretical relationships. Five

indicators of transportability and three indicators of narrative transportation were eliminated from analysis since they displayed outer loadings below .6. Additional indicators had loadings between .6 and .7, but we retained them due to the explorative nature of our study [53], and because all constructs' composite reliability and Cronbach's Alpha scores were above the recommended threshold of .7 [53, 54]. The heterotrait-monotrait ratio of correlations of enjoyment and narrative transportation was .859, and, thus, slightly above the recommended threshold of .85 [55]. However, we decided to retain both constructs since the average variance extracted (AVE) for each construct was greater than the suggested minimum of .5 and the square root of each construct's AVE exceeded the inter-construct correlations, demonstrating adequate discriminant validity [56] (see Table 1). Furthermore, all items loaded highest on their theoretical construct. One-way ANOVAs revealed significant differences between the high group means and the low group means for content-task congruence ($M_{\text{high}} = 5.98$ vs. $M_{\text{low}} = 3.93$; $F(1,324) = 128.750$, $p < .001$) as well as discourse-task congruence ($M_{\text{high}} = 5.70$ vs. $M_{\text{low}} = 3.47$; $F(1,324) = 152.979$, $p < .001$).

To assess our research model, we adopted PLS-SEM and used the SmartPLS software, version 4.0.8.7. We tested the structural model by evaluating the direct effects and the explained variances (R^2). While assessing the model, we controlled for *home workout frequency*, *home workout app usage*, *age*, *physical activity enjoyment*, *fitness level*, and *audio book use* (see Figure 3). The results show a positive, but not significant effect of content-task congruence on narrative transportation (.098; $p = .183$), thus **not supporting H1a**, but a positive significant effect of discourse-task congruence on narrative transportation (.459; $p < .001$), thus **supporting H1b**. The interaction effects of transportability on narrative transportation

Table 1: Construct Reliability and Correlations

Construct	CR	AVE	Inter-construct correlations											
			1)	2)	3)	4)	5)	6)	7)	8)	9)	10)	11)	12)
1) Age	1	1	1											
2) Audiobook Use	1	1	-.308	1										
3) Content-Task Congruence	.973	.897	-.046	.183	.947									
4) Discourse-Task Congruence	.971	.888	-.103	.186	.771	.943								
5) System Use Enjoyment	.964	.933	-.050	.214	.543	.639	.966							
6) Fitness Level	1	1	-.018	-.100	-.025	-.019	-.009	1						
7) Home Workout App Usage	1	1	-.266	.391	.069	.098	.105	-.207	1					
8) Home Workout Behavior	1	1	-.028	.207	.065	.044	.171	-.372	.343	1				
9) Instrumental Task Outcome	.943	.807	.008	.084	.232	.289	.400	-.156	.081	.275	.898			
10) Narrative Transportation	.937	.670	-.121	.315	.534	.619	.820	-.050	.111	.153	.391	.818		
11) Physical Activity Enjoyment	.947	.859	.008	.114	.212	.224	.350	-.376	.162	.306	.366	.347	.927	
12) Transportability	.956	.608	-.216	.419	.269	.269	.393	-.081	.199	.139	.255	.553	.282	.780

**Figure 3: Structural Model**

for content-task congruence (.129; $p = .130$) and discourse-task congruence (-.129; $p = .120$) were not significant, indicating **no support for H2a and H2b**. The effect of narrative transportation on enjoyment was significantly positive (.820; $p < .001$), **supporting H3**. The effect of narrative transportation on instrumental task outcome (.158; $p = .096$) was positive, but not significant in the control model, indicating **no support for H4**. However, effect was significant in the model without control variables (see A.4), which leaves room for further investigations. Finally, we found a positive significant effect of enjoyment on instrumental task outcome (.179; $p < .05$), indicating **support for H5**. In terms of R^2 , our model explains 56% of narrative transportation, 67% of enjoyment, and 25% of instrumental task outcome. Additionally, we found a positive significant effect of transportability on narrative transportation as well as positive significant effects of home workout frequency and physical activity enjoyment on instrumental task outcome.

6 DISCUSSION

This late breaking work constitutes a first attempt to better understand the mental state of narrative transportation within users of gamified IS. To explain under which circumstances narrative transportation emerges (RQ2), we conceptually distinguished and empirically investigated two types of narrative-task congruence (i.e., content-task congruence and discourse-task congruence). In our experiment, higher levels of discourse-task congruence led to higher levels of narrative transportation within users of gamified IS. Yet, we could not find support for a similar effect of content-task congruence on narrative transportation. This was surprising for us since the statements obtained through open-ended questions (see A.5) indicated that some participants in the low content-task congruence groups felt confused about linking the events described in the narrative to the instrumental task of running or sprinting in place. It is possible that users of gamified IS are more likely

to accept low levels of content-task congruence when the narrative discourse is congruent with the instrumental task. Regarding the effect of narrative transportation on meaningful engagement (RQ1), our results also yield interesting insights. For one, the results support our hypothesis that experiencing narrative transportation has a positive influence on system use enjoyment. We attribute this mainly to the fact that individuals generally enjoy temporarily leaving reality behind and immersing themselves in narrative worlds, which keeps their minds from consciously thinking about the instrumental task [10]. For another, we found that system use enjoyment has a positive influence on physical activity as an instrumental outcome, which supports our assumption that for users of gamified IS experiential outcomes can yield important value in contributing to instrumental outcomes. Finally, the direct effect of narrative transportation on instrumental task outcome was positive as hypothesized yet not significant in the control model, but slightly below the threshold of .05 in the model without control variables. This highlights the need for further theory-driven research in the lab or in the field to better understand this relationship and identify additional influential factors.

Our work contributes to research concerned with the questions how and under which circumstances individual game design elements fit intended instrumental task outcomes of gamified IS (i.e., task congruence principle of gamification design [1]). The results of our study reaffirm that to foster meaningful engagement, narratives in gamified IS may benefit from being compatible with the targeted instrumental task. Our study also contributes to gamification research by shedding light on some of the psychological and behavioral outcomes of narratives as a game design element, thereby answering an often-raised call for moving gamification research beyond traditional elements such as points, badges, and leaderboards [2]. Regarding practice, our results provide support for the general assumption that narratives are a useful tool in evoking meaningful engagement within users of gamified IS. In particular, such narratives that enable users to experience high levels of narrative transportation can be a powerful tool for designers of gamified IS in their quest to create fun and joyful use experiences that also yield potential in increasing instrumental task outcomes [2]. The results of our study also reinforce the need to move toward context-specific gamification solutions. Designers of gamified IS that aim to utilize the persuasive potential of narratives should pay attention to adjust their narratives to fit the instrumental task that they aim to support with their system. Only then will they be able to unfold the full potential of narratives to positively influence meaningful engagement.

This study is limited by several factors. First, due to the novelty of the topic of this research we used self-developed items for assessing content-task congruence and discourse-task congruence. As the topic gains in relevance, future research may engage in the rigorous development of established scales for the two dimensions of narrative-task congruence. Such research will also help in empirically validating to which extent content and discourse are perceived separately by users of gamified IS or whether they become inseparable once they manifest in narrative objects. Furthermore, we used an online experiment with self-reported measures for instrumental task outcome. While we aimed to address this issue by deploying established guidelines for quality control of online experiments

and excluded participants based on answers to open questions, we cannot rule out the possibility that some participants did not sufficiently engage in the physical activity or misjudged their own performance, which could also be one reason for the inconclusive findings on instrumental task outcomes. Future research could overcome this limitation by conducting lab or field experiments with more objective measures of instrumental task outcome (e.g., using sensory data on a mobile or wearable device to calculate physical activity behavior). Lastly, our study only investigated the effects of narratives in gamified IS for physical activity. Future research should expand toward other application contexts of gamified IS to investigate the role of narrative transportation in other contexts of gamification as well.

7 CONCLUSION

To sum up, the results of our study indicate that congruence between narrative and instrumental task can influence the emergence of narrative transportation within users of gamified IS and that narrative transportation positively influences the experiential outcome of system use enjoyment. Furthermore, we find that users who enjoy using a gamified IS that utilizes narratives tend to have higher instrumental task outcomes from using the system than others. This late breaking work should be understood as an exploratory attempt to study the role of narrative transportation and narrative-task congruence in gamified IS. It yields promising initial insights into the effects of narratives as game design elements in gamified IS but also illustrates the relevance for further research especially with regard to the operationalization of narrative-task congruence. Our study aims to open the academic discussion on narrative transportation and its antecedents in gamified IS thereby strengthening theory-driven gamification research beyond points, badges, and leaderboards.

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